IPW3 : ONERAM6 Code-to-Code Comparisons

Karim Zayni, PhD Student
Maxime Blanchet, PhD Student
Éric Laurendeau, PhD, Professor

September 22, 2026

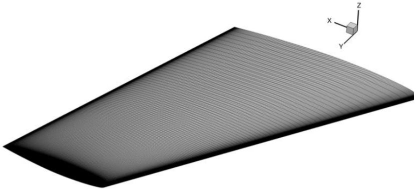
POLYTECHNIQUE MONTRÉAL
TECHNOLOGICAL UNIVERSITY

Table of Contents

IPW3 – Participants

ID	Participant / Organization
001	CIRA
002	AeroTex
003	NRC
004	ONERA
006	Dassault
007	Polytechnique Montréal
008	Collins Aerospace
009	Sikorsky
010	NASA
013	Boeing
014	Airbus
015	NIAR
019	Synopsys
020	Bombardier

Geometry and Aerodynamic Conditions



ONERA M6 L1 Surface Mesh

Geometry

Parameter	Value
Wing	ONERA M6
Root chord [m]	1.0
Semi-span [m]	1.4760
Mean aerodynamic chord [m]	0.801673
Reference area [m^2]	1.1532
Angle of attack, α [°]	2.0

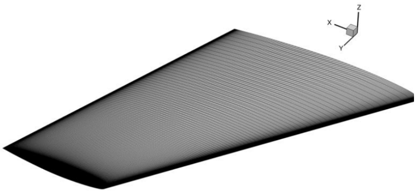
Aerodynamic Conditions

Parameter	Value
M_∞ [-]	0.40
U_∞ [m/s]	131.064
T_∞ [K]	267.15
ρ_∞ [kg/m^3]	0.68301
Re_c [-]	4.2558×10^6

Quantities of Interest

$$C_L, \quad C_D, \quad C_M, \quad C_p$$

Icing Conditions



ONERA M6 L1 Surface Mesh

Grid	Nodes
L1	60.78 M
L2	7.63 M
L3	0.960 M
L4	0.122 M

Icing Conditions

Parameter	Value
T_∞ [K]	267.15
T_0 [K]	275.7
M_∞	0.40
U_∞ [m/s]	131.064
LWC [g/m^3]	0.51
MVD [μm]	20
Exposure time [s]	1350

Droplet Distribution : 1, 3, 7 & 15 bin distributions are considered

Common Modeling Parameters

A dimensional roughness of $k_s = 1.0$ **mm** is prescribed as the baseline, and a common ice density of $\rho_{\text{ice}} = 680\ \text{kg/m}^3$ is used for the comparison.

Quantities of Interest

$$\beta, \quad HTC, \quad T_s, \quad FF, \quad m_{\text{water}}, \quad m_{\text{ice}}, \quad m_{\text{evap}}$$

ONERA M6 – Modeling Choices Across Submissions

- **Turbulence:** k–ω formulations were dominant (**80%**), with SA used by **20%**.
- **Droplet formulation:** Predominantly Eulerian (**80%**), with **20%** Lagrangian.
- **Heat transfer:** Documented approaches included recovery-temperature and boundary-layer integral methods.
- **Thermodynamics:** Messinger-based models were dominant (**80%**), with a SWIM-based approach also represented.
- **Geometry evolution:** Most commonly based on Lagrangian/IB/node displacement; level-set and interpolation-based approaches were also reported.
- **Ice accretion strategy:** Both single-shot (**4 submissions**) and multilayer/multishot (**3 submissions**) approaches were represented.

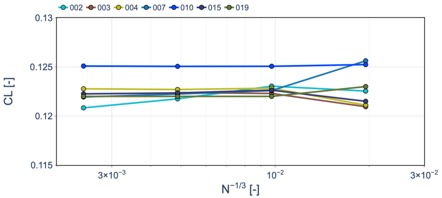
Modeling choices disclosed by participants: 001, 002, 004, 007, 008, 009, 014, and 019.

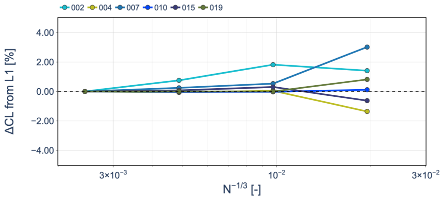
ONERA M6 – Summary of Data Received

ID	L1 1	L1 3	L1 7	L1 15	L2 1	L2 3	L2 7	L2 15	L3 1	L3 3	L3 7	L3 15	L4 1	L4 3	L4 7	L4 15
001	✓				✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
002	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
003						*				*				*		
004	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
006																
007	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
008																
009																
010	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
013																
014																
015	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
019	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
020																
Total	**7**	**6**	**6**	**6**	**7**	**7**	**7**	**7**	**7**	**7**	**7**	**7**	**7**	**7**	**7**	**7**

✓: populated CutData received for the corresponding droplet-bin distribution.
∗: Excel results only, no CutData slices.

ONERA M6 – Lift Coefficient Grid Convergence

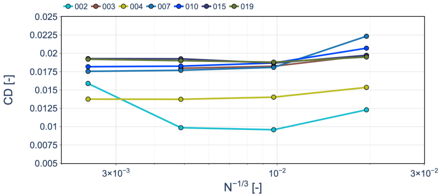


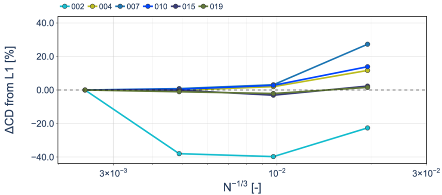


Assessment

- At L1, the median lift coefficient is $C_L = 0.12213$, with an IQR of 6.89×10^{-4} (0.56%).
- Coarse-grid deviations generally remain within a few percent of the corresponding L1 predictions.
- The code-to-code spread in C_L remains small across the grid levels.

ONERA M6 – Drag Coefficient Grid Convergence

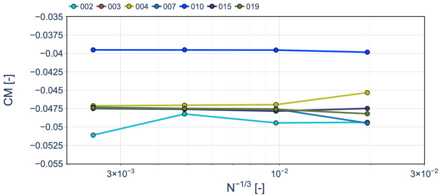


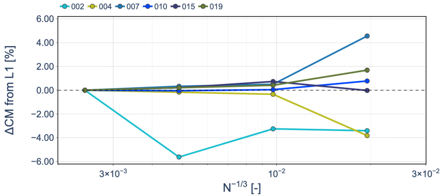


Assessment

- At L1, the median drag coefficient is $C_D = 0.01786$ (178.6 drag counts), with an IQR of 2.64×10^{-3} (26.4 drag counts, 14.8%).
- C_D exhibits considerably larger code-to-code variability than C_L.
- Several submissions show non-monotonic sensitivity to spatial grid refinement.
- Relative deviations from the corresponding L1 predictions reach approximately 40% for selected submissions.
- Flow-convergence difficulties were reported for some submissions and may contribute to the observed grid sensitivity and variability in C_D.

ONERA M6 – Pitching Moment Grid Convergence

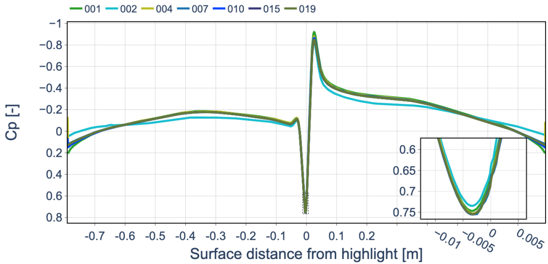




Assessment

- At L1, the median pitching-moment coefficient is $C_M = -0.04736$, with an IQR of 2.93×10^{-4} (0.62%).
- Most submissions exhibit limited sensitivity to spatial grid refinement.

ONERA M6 – Surface Pressure Distribution (L1)

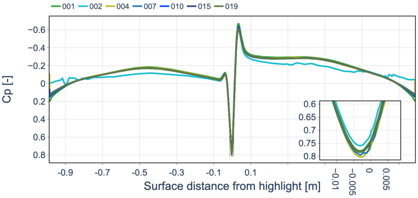


$y = 0.75$ m

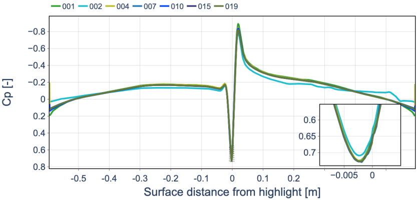
Assessment

- The submitted C_p distributions are generally closely clustered across all three spanwise locations.
- The characteristic pressure variations associated with the leading-edge and suction-side peak are consistently captured by the participants.
- Overall agreement remains similar from the inboard to the outboard spanwise stations.

ONERA M6 – Surface Pressure Distribution (L1)



$y = 0.10$ m



$y = 1.40$ m

Assessment

- The submitted C_p distributions are generally closely clustered across all three spanwise locations.
- The characteristic pressure variations associated with the leading-edge and suction-side peak are consistently captured by the participants.
- Overall agreement remains similar from the inboard to the outboard spanwise stations.

ONERA M6 – Aerodynamic Assessment Summary

Grid Convergence

- C_L: very weak grid sensitivity, with closely clustered predictions across the grid hierarchy.
- C_D: substantially greater grid sensitivity and code-to-code variability, including non-monotonic behavior for selected submissions.
- C_M: limited within-code grid sensitivity for most submissions.

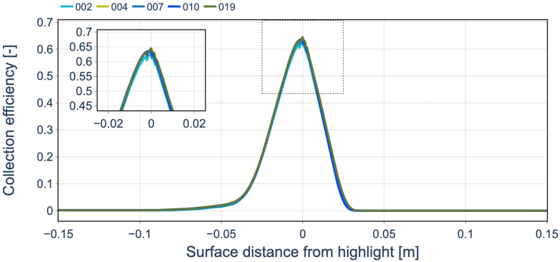
Code-to-Code Variability at L1

	Median	IQR	Relative IQR
C_L	0.12213	6.89×10^{-4}	0.56%
C_D	0.01786	2.64×10^{-3}	14.8%
C_M	−0.04736	2.93×10^{-4}	0.62%

Aerodynamic Takeaway

The lift and pitching-moment predictions exhibit weak grid sensitivity and relatively tight code-to-code agreement for most submissions. In contrast, drag remains considerably more sensitive to both numerical implementation and spatial resolution. The C_p distributions are nevertheless consistently reproduced across the three spanwise stations.

ONERA M6 – Collection Efficiency (15 Bins, L1)

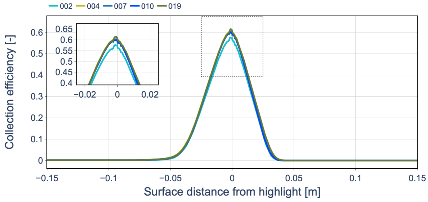


$y = 0.75$ **m**

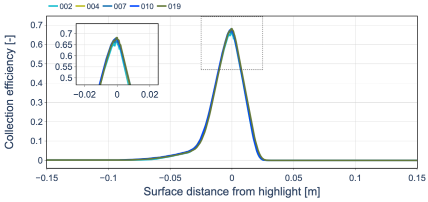
Assessment

- Good code-to-code agreement is observed in the collection-efficiency distribution at the representative middle spanwise station ($y = 0.75$ m).
- The peak magnitude and location are closely clustered across submissions.
- Similar agreement is observed at the inboard and outboard stations.
- Small oscillations near the peak are likely associated with the surface discretization in the highly curved leading-edge region.

ONERA M6 – Collection Efficiency (15 Bins, L1)



$y = 0.10$ **m**

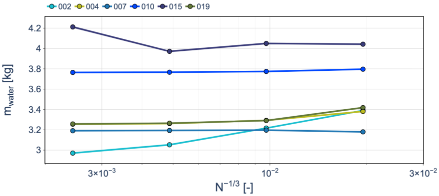


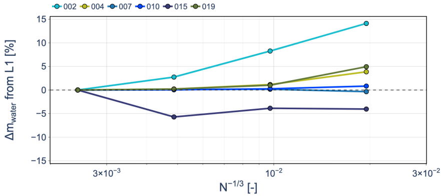
$y = 1.40$ **m**

Assessment

- Good code-to-code agreement is observed in the collection-efficiency distribution at the representative middle spanwise station ($y = 0.75$ m).
- The peak magnitude and location are closely clustered across submissions.
- Similar agreement is observed at the inboard and outboard stations.
- Small oscillations near the peak are likely associated with the surface discretization in the highly curved leading-edge region.

ONERA M6 – Water Mass Grid Convergence (15 Bins)

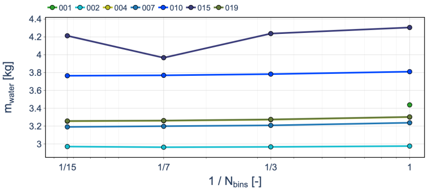




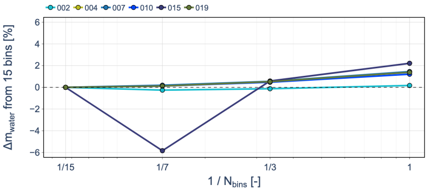
Assessment

- For the 15-bin distribution, the L1 median water mass is 3.257 kg, with an IQR of 0.431 kg (13.2%).
- Most submissions exhibit limited sensitivity to spatial grid refinement, with changes generally remaining within approximately 5% of their corresponding L1 predictions.
- Code-to-code differences in total water mass are larger than the grid-refinement effect for most submissions.

ONERA M6 – Droplet Distribution Sensitivity (L1)



Impinged Water Mass

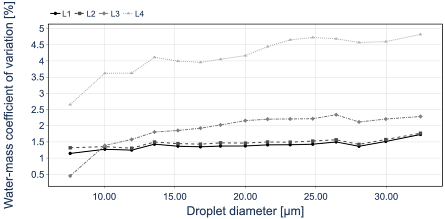


Relative to 15 Bins

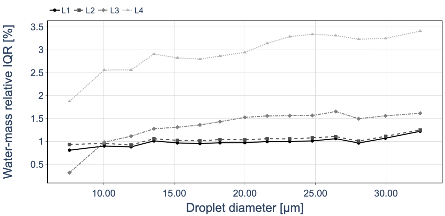
Assessment

- All the distributions produce very similar impinged water masses for most submissions.
- Relative differences remain within approximately 2% for most submissions and distributions, with one isolated 7-bin deviation of approximately 6%.
- The narrow droplet-size range (6.16–33.84 μm around an MVD of 20 μm) contributes to the weak distribution sensitivity.

ONERA M6 – Water-Mass Variability by Droplet Diameter



Coefficient of Variation

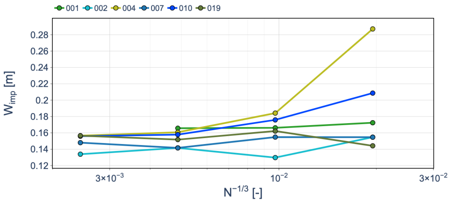


Relative IQR

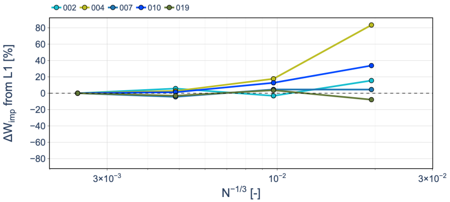
Assessment

- On the fine L1 and L2 grids, the code-to-code variability remains low across the entire droplet-diameter range, with relative IQR values generally around 1%.
- Variability increases on the coarser L3 and L4 grids, indicating a stronger grid-resolution effect than droplet-size dependence.
- No pronounced dependence on droplet diameter is observed over the relatively narrow 6.16–33.84 μm range, centered around an MVD of 20 μm.
- This relatively narrow droplet-size distribution also helps explain the weak sensitivity of the integrated impinged water mass to the number of bins.

ONERA M6 – Impingement Width Grid Convergence (15 Bins)



$y = 0.75$ **m**

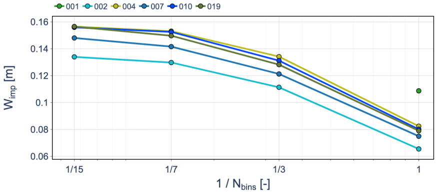


Relative to L1

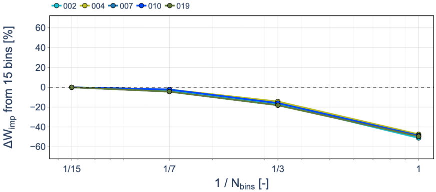
Assessment

- Due to the oscillations observed near the β peak, the peak magnitude and position are excluded from the grid-convergence assessment; the impingement width is retained as the primary profile-based metric.
- The impingement width is defined from the lower and upper $\beta = 0.001$ threshold crossings.
- Most submissions show comparatively limited grid sensitivity between L1 and L2.
- Larger and non-monotonic deviations are observed on the coarser grids for several submissions, indicating greater sensitivity of the predicted impingement limits to spatial resolution.

ONERA M6 – Impingement Width Distribution Sensitivity (L1)



$y = 0.75$ **m**



Relative to 15 Bins

Assessment

- The predicted impingement width systematically increases as the droplet distribution is refined from 1 to 15 bins.
- A noticeable increase remains between the 7- and 15-bin distributions, indicating that convergence with respect to droplet-distribution resolution has not yet been clearly demonstrated.
- In contrast, the integrated impinged water mass varies by only approximately 2% between the single- and 15-bin distributions for most submissions.
- Droplet-distribution resolution therefore has a substantially stronger influence on the **spatial extent** of impingement than on the integrated collected water mass.

ONERA M6 – Droplet Impingement Summary

Spatial Grid Convergence

- Collection-efficiency distributions show good code-to-code agreement across the three spanwise stations.
- Integrated water mass exhibits limited grid sensitivity for most submissions, while code-to-code differences remain larger.
- Impingement limits are more sensitive to spatial resolution, particularly on the coarser grids.

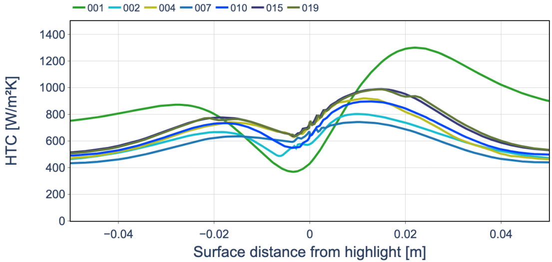
Droplet-Size Distribution

- Integrated water mass is weakly sensitive to droplet discretization; even the single-MVD approximation remains within approximately 2% of the 15-bin prediction for most submissions.
- In contrast, impingement width systematically increases from 1 to 15 bins, with convergence not yet clearly demonstrated at 15 bins.
- The relatively narrow 6.16–33.84 μm range around the 20 μm MVD contributes to the weak sensitivity of the integrated water mass.

Impingement Takeaway

Integrated water mass can appear converged while the spatial impingement extent remains sensitive to both spatial and droplet-distribution resolution.

ONERA M6 – Convective Heat-Transfer Distribution (L1)

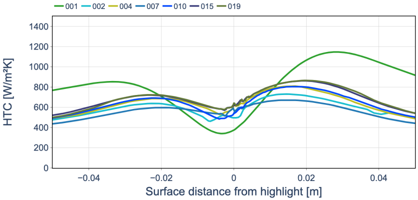


$y = 0.75$ **m**

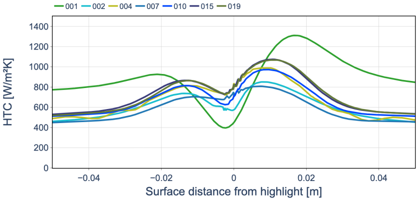
Assessment

- At the prescribed 1 mm roughness, considerable code-to-code differences are observed in *HTC*, particularly near the leading edge.
- Similar variability is observed at the other spanwise stations, indicating that roughness height alone does not explain the code-to-code scatter; differences in the *HTC* computation methodology also contribute.
- When the roughness height is varied, increasing roughness systematically increases the predicted *HTC*.

ONERA M6 – Convective Heat-Transfer Distribution (L1)



$y = 0.10$ **m**

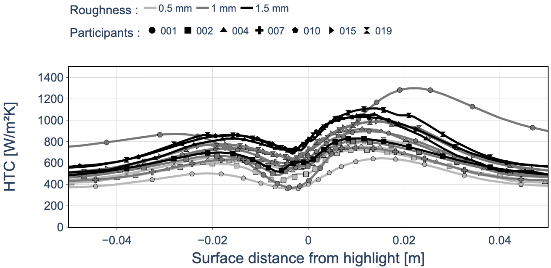


$y = 1.40$ **m**

Assessment

- At the prescribed 1 mm roughness, considerable code-to-code differences are observed in *HTC*, particularly near the leading edge.
- Similar variability is observed at the other spanwise stations, indicating that roughness height alone does not explain the code-to-code scatter; differences in the *HTC* computation methodology also contribute.
- When the roughness height is varied, increasing roughness systematically increases the predicted *HTC*.

ONERA M6 – Convective Heat-Transfer Distribution (L1)

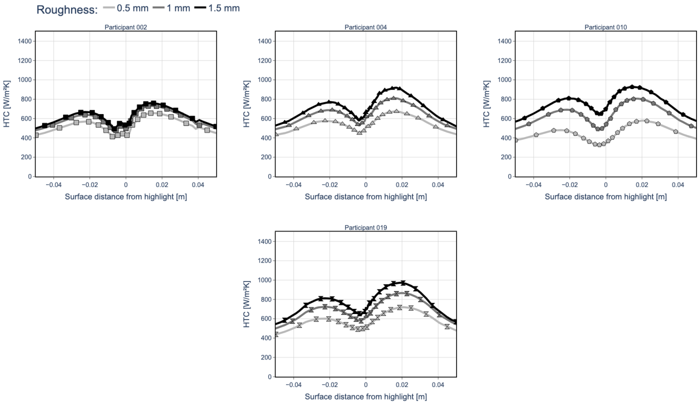


$y = 0.75$ **m – Roughness Sensitivity**

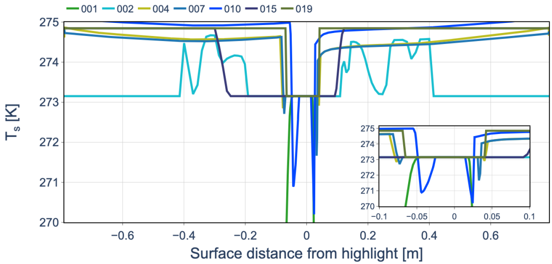
Assessment

- At the prescribed 1 mm roughness, considerable code-to-code differences are observed in *HTC*, particularly near the leading edge.
- Similar variability is observed at the other spanwise stations, indicating that roughness height alone does not explain the code-to-code scatter; differences in the *HTC* computation methodology also contribute.
- When the roughness height is varied, increasing roughness systematically increases the predicted *HTC*.

ONERA M6 – Participant-Level HTC Roughness Sensitivity (L1)



ONERA M6 – Surface Temperature Distribution (L1)

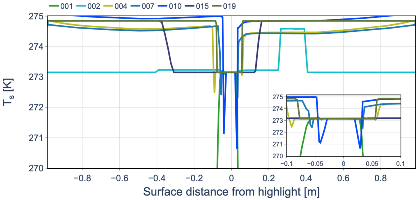


$y = 0.75$ m – **Roughness: 1 mm**

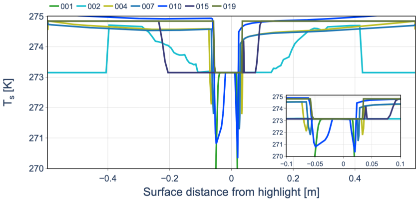
Assessment

- Most submissions predict surface temperatures close to the freezing temperature over a large portion of the surface.
- Local code-to-code differences are concentrated near the leading edge.
- Similar differences are observed at the inboard and outboard spanwise stations.

ONERA M6 – Surface Temperature Distribution (L1)



$y = 0.10$ **m**

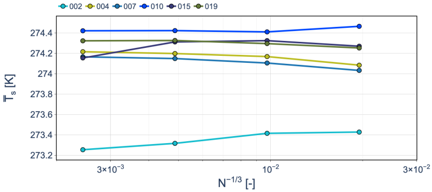


$y = 1.40$ **m**

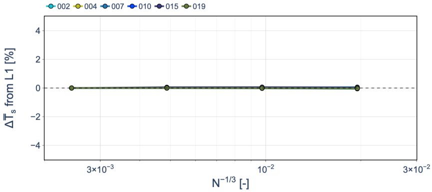
Assessment

- Most submissions predict surface temperatures close to the freezing temperature over a large portion of the surface.
- Local code-to-code differences are concentrated near the leading edge.
- Similar differences are observed at the inboard and outboard spanwise stations.

ONERA M6 – Mean Surface Temperature Grid Convergence



Mean Surface Temperature



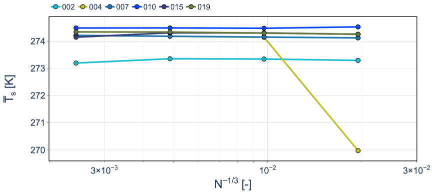
Relative to L1

$y = 0.75$ m **– Roughness: 1 mm**

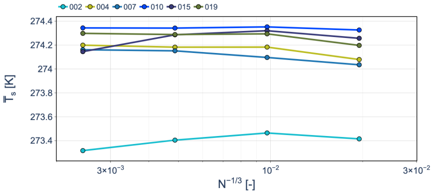
Assessment

- Mean surface temperature exhibits very limited grid sensitivity, with differences relative to L1 remaining well below 1% for most submissions.
- Most submissions are closely clustered.
- Similar grid-convergence behavior is observed at the inboard and outboard spanwise stations.
- Increasing the prescribed roughness height introduces a slight difference in the mean surface temperature.

ONERA M6 – Mean Surface Temperature Grid Convergence



$y = 0.10$ **m**

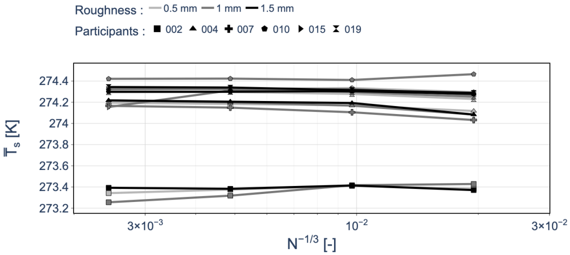


$y = 1.40$ **m**

Assessment

- Mean surface temperature exhibits very limited grid sensitivity, with differences relative to L1 remaining well below 1% for most submissions.
- Most submissions are closely clustered.
- Similar grid-convergence behavior is observed at the inboard and outboard spanwise stations.
- Increasing the prescribed roughness height introduces a slight difference in the mean surface temperature.

ONERA M6 – Mean Surface Temperature Grid Convergence

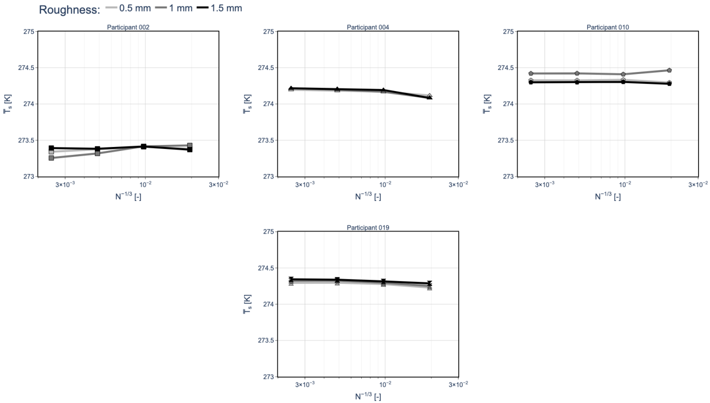


$y = 0.75$ **m – Roughness Sensitivity**

Assessment

- Mean surface temperature exhibits very limited grid sensitivity, with differences relative to L1 remaining well below 1% for most submissions.
- Most submissions are closely clustered.
- Similar grid-convergence behavior is observed at the inboard and outboard spanwise stations.
- Increasing the prescribed roughness height introduces a slight difference in the mean surface temperature.

ONERA M6 – Participant-Level Mean Surface Temperature Roughness Sensitivity (L1)



ONERA M6 – Integrated Convective Heat Transfer

Grid-Convergence Metric

To assess whether the differences in the local *HTC* distributions translate into a clear grid-convergence behavior, an integrated convective heat-transfer quantity is evaluated along the surface slice:

$$\boxed{Q'_c = \int HTC(s)\,[T_s(s) - T_{\mathrm{rec}}(s)]\,ds} \qquad [Q'_c] = \mathrm{W/m}.$$

The submitted *HTC* and surface temperature T_s are used directly. The recovery temperature T_{rec}, however, was not part of the submitted dataset and must therefore be reconstructed.

Recovery Temperature Reconstruction

The submitted pressure coefficient is used to estimate the local external-flow state and subsequently the recovery temperature:

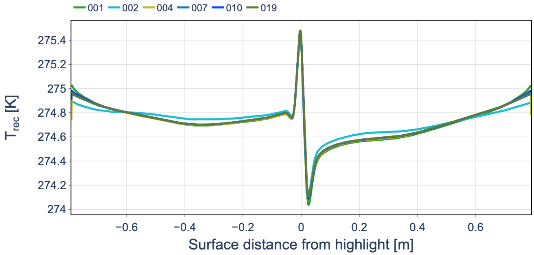
$$\boxed{C_p(s) \longrightarrow p_e(s) \longrightarrow M_e(s) \longrightarrow T_e(s) \longrightarrow T_{\mathrm{rec}}(s)}$$

Assuming $p_s \simeq p_e$ and an isentropic external flow,

$$\frac{p_e}{p_\infty} = 1 + \frac{\gamma}{2} M_\infty^2 C_{p,e}, \qquad M_e^2 = \frac{2}{\gamma - 1}\left[\left(\frac{p_0}{p_e}\right)^{(\gamma-1)/\gamma} - 1\right],$$

$$T_e = \frac{T_0}{1 + \dfrac{\gamma - 1}{2} M_e^2}, \qquad T_{\mathrm{rec}} = T_e\left[1 + r\frac{\gamma - 1}{2} M_e^2\right], \qquad r = Pr^{1/3} \approx 0.9.$$

ONERA M6 – Reconstructed Recovery Temperature Distribution (L1)

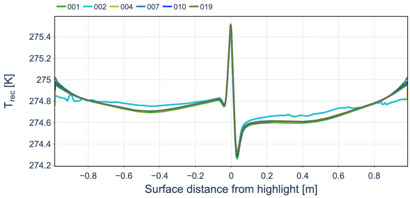


$y = 0.75$ **m – Roughness: 1 mm**

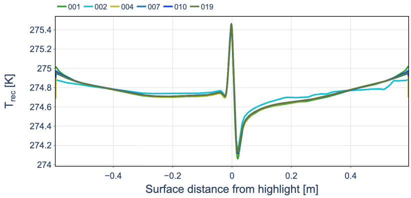
Assessment

- The reconstructed recovery-temperature distributions are closely aligned across submissions.
- This agreement is consistent with the good code-to-code agreement previously observed for C_p, from which T_{rec} is reconstructed.
- Similar agreement is observed at the inboard and outboard spanwise stations.

ONERA M6 – Reconstructed Recovery Temperature Distribution (L1)



$y = 0.10$ **m**

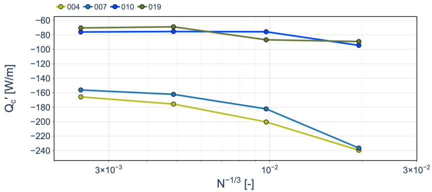


$y = 1.40$ **m**

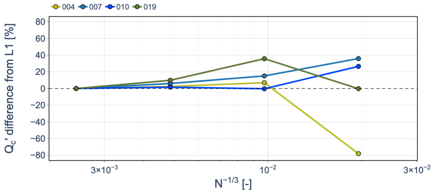
Assessment

- The reconstructed recovery-temperature distributions are closely aligned across submissions.
- This agreement is consistent with the good code-to-code agreement previously observed for C_p, from which T_{rec} is reconstructed.
- Similar agreement is observed at the inboard and outboard spanwise stations.

ONERA M6 – Convective Heat Transfer Grid Convergence



Q_c'



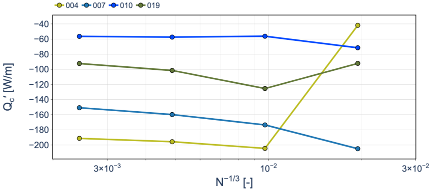
Relative to L1

$y = 0.75$ m – **Roughness: 1 mm**

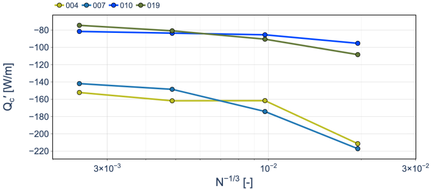
Assessment

- Considerable code-to-code variability is observed in Q_c' at $y = 0.75$ m, with an L1 median of -116.1 W/m and an IQR of 83.9 W/m.
- Relative to L1, most submissions show comparatively limited changes between L1 and L2, while deviations of approximately 25–50% develop on the coarser grids for selected submissions.
- Large L1 code-to-code variability persists across the span, with IQR values of 77.5, 83.9, and 64.6 W/m at $y = 0.10$, 0.75, and 1.40 m, respectively.
- Surface roughness introduces a systematic effect on Q_c'.

ONERA M6 – Convective Heat Transfer Grid Convergence



$y = 0.10$ **m**

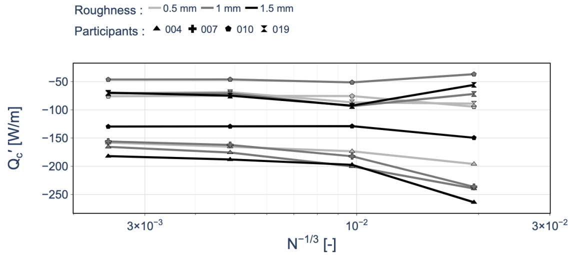


$y = 1.40$ **m**

Assessment

- Considerable code-to-code variability is observed in Q_c' at $y = 0.75$ m, with an L1 median of -116.1 W/m and an IQR of 83.9 W/m.
- Relative to L1, most submissions show comparatively limited changes between L1 and L2, while deviations of approximately 25–50% develop on the coarser grids for selected submissions.
- Large L1 code-to-code variability persists across the span, with IQR values of 77.5, 83.9, and 64.6 W/m at $y = 0.10$, 0.75, and 1.40 m, respectively.
- Surface roughness introduces a systematic effect on Q_c'.

ONERA M6 – Convective Heat Transfer Grid Convergence

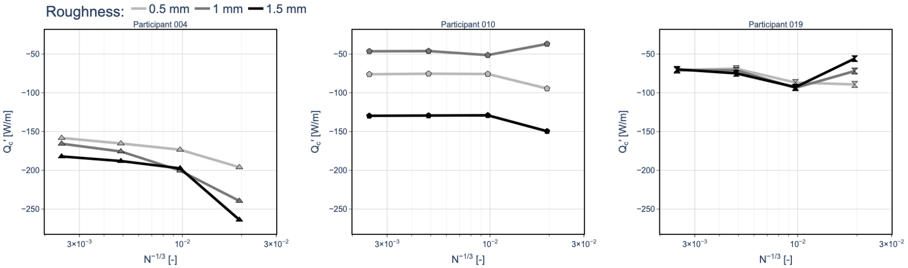


$y = 0.75$ **m – Roughness Sensitivity**

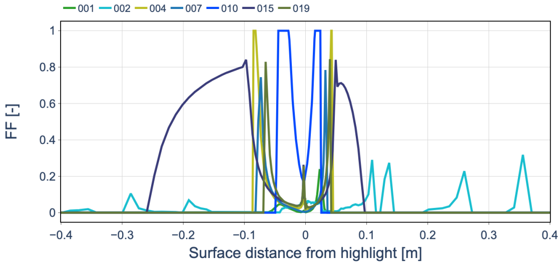
Assessment

- Considerable code-to-code variability is observed in Q_c' at $y = 0.75$ m, with an L1 median of -116.1 W/m and an IQR of 83.9 W/m.
- Relative to L1, most submissions show comparatively limited changes between L1 and L2, while deviations of approximately 25–50% develop on the coarser grids for selected submissions.
- Large L1 code-to-code variability persists across the span, with IQR values of 77.5, 83.9, and 64.6 W/m at $y = 0.10$, 0.75, and 1.40 m, respectively.
- Surface roughness introduces a systematic effect on Q_c'.

ONERA M6 – Participant-Level HTC Roughness Sensitivity (L1)



ONERA M6 – Freezing-Fraction Distribution (L1)

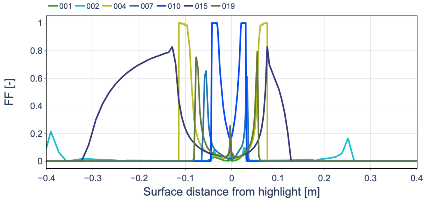


$y = 0.75$ **m – Roughness: 1 mm**

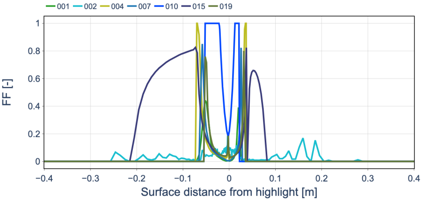
Assessment

- Large code-to-code differences are observed in both the magnitude and spatial extent of the predicted freezing fraction.
- Substantial differences in the predicted freezing regime persist across the inboard and outboard spanwise stations.

ONERA M6 – Freezing-Fraction Distribution (L1)



$y = 0.10$ **m**

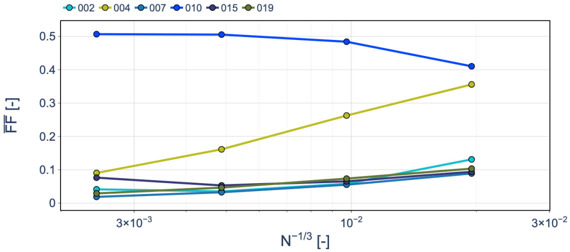


$y = 1.40$ **m**

Assessment

- Large code-to-code differences are observed in both the magnitude and spatial extent of the predicted freezing fraction.
- Substantial differences in the predicted freezing regime persist across the inboard and outboard spanwise stations.

ONERA M6 – Mean Freezing Fraction Grid Convergence

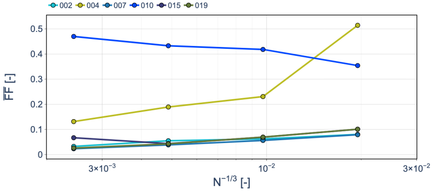


$y = 0.75$ **m – Roughness: 1 mm**

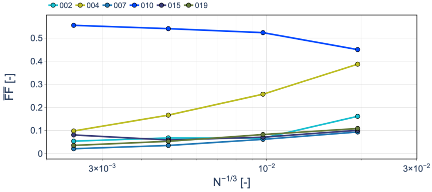
Assessment

- Mean freezing fraction exhibits very large code-to-code variability, despite the comparatively close mean surface-temperature predictions for most submissions.
- Grid sensitivity is submission-dependent, with some predictions remaining relatively stable while others vary substantially across the grid hierarchy.
- Large code-to-code differences persist at the inboard and outboard spanwise stations.
- Prescribed surface roughness introduces an additional systematic effect.

ONERA M6 – Mean Freezing Fraction Grid Convergence



$y = 0.10$ **m**

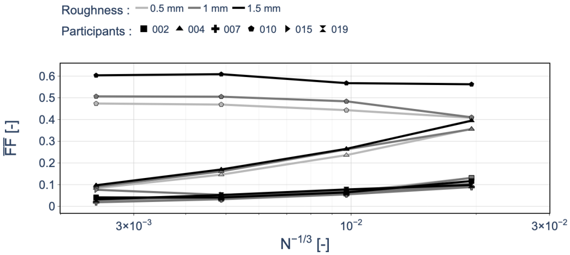


$y = 1.40$ **m**

Assessment

- Mean freezing fraction exhibits very large code-to-code variability, despite the comparatively close mean surface-temperature predictions for most submissions.
- Grid sensitivity is submission-dependent, with some predictions remaining relatively stable while others vary substantially across the grid hierarchy.
- Large code-to-code differences persist at the inboard and outboard spanwise stations.
- Prescribed surface roughness introduces an additional systematic effect.

ONERA M6 – Mean Freezing Fraction Grid Convergence

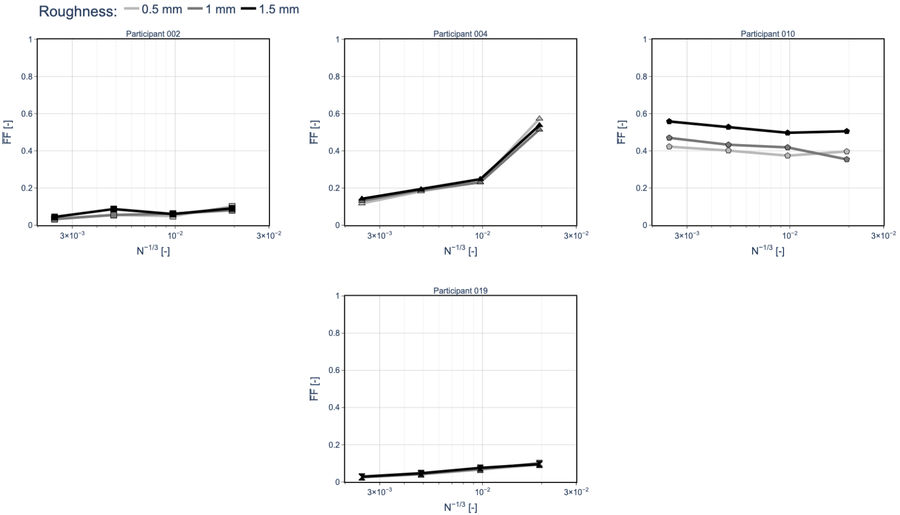


$y = 0.75$ **m – Roughness Sensitivity**

Assessment

- Mean freezing fraction exhibits very large code-to-code variability, despite the comparatively close mean surface-temperature predictions for most submissions.
- Grid sensitivity is submission-dependent, with some predictions remaining relatively stable while others vary substantially across the grid hierarchy.
- Large code-to-code differences persist at the inboard and outboard spanwise stations.
- Prescribed surface roughness introduces an additional systematic effect.

ONERA M6 – Participant-Level Freezing Fraction Roughness Sensitivity (L1)



ONERA M6 – Convective Heat Transfer & Thermodynamics Summary

Convective Heat Transfer

- Considerable code-to-code variability is observed in HTC, while reconstructed T_{rec} distributions remain closely aligned.
- Consequently, Q_c' exhibits substantial code-to-code variability, with an L1 IQR of 83.9 W/m at $y = 0.75$ m.

Surface Thermodynamics

- Mean surface temperature shows limited grid sensitivity and comparatively close agreement for most submissions.
- In contrast, freezing fraction exhibits large code-to-code variability in both magnitude and spatial extent.

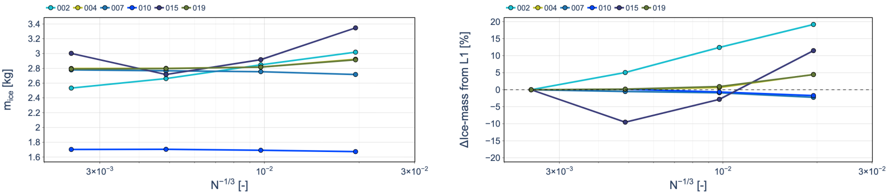
Roughness Sensitivity

- Increasing roughness systematically increases HTC and mean surface temperature.
- Roughness also affects Q_c' and freezing fraction, but does not explain the large overall code-to-code spread.

Thermodynamics Takeaway

Despite consistent aerodynamic loading and T_{rec}, substantial variability develops through the convective and freezing models, with the largest differences appearing in the predicted freezing regime.

ONERA M6 – Ice Mass Grid Convergence (15 Bins)



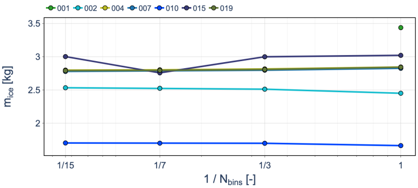
Ice Mass **Relative to L1**

Roughness: 1 mm

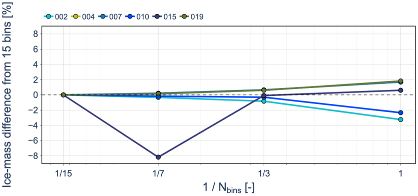
Assessment

- Ice-mass grid sensitivity is strongly submission-dependent.
- Relative to L1, coarse-grid differences remain below approximately 5% for several submissions, but reach approximately 20% and 11% for selected submissions on L4.

ONERA M6 – Ice Mass Sensitivity to Droplet Distribution



Ice Mass



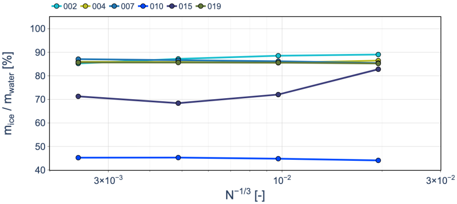
Relative to 15 Bins

L1 – Roughness: 1 mm

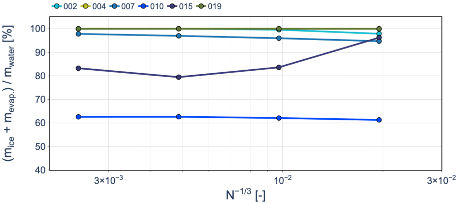
Assessment

- Ice mass exhibits limited sensitivity to droplet-distribution resolution for most submissions.
- Relative differences from the 15-bin prediction generally remain within approximately 3%, with one isolated 7-bin deviation of approximately 8%.
- Code-to-code differences remain considerably larger than the droplet-distribution effect.
- This behavior is consistent with the weak distribution sensitivity previously observed for the integrated impinged water mass.

ONERA M6 – Ice and Evaporation Mass Ratios



Ice / Water

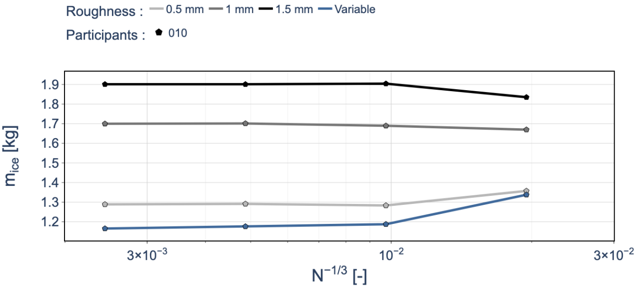


(Ice + Evaporation) / Water

Assessment

- Most submissions convert approximately 85–89% of the impinged water mass into ice.
- Accounting for evaporation brings the mass balance close to 100% for some submissions.
- The remaining mass deficit for some submissions may be associated with water shedding from the surface or other mechanisms.

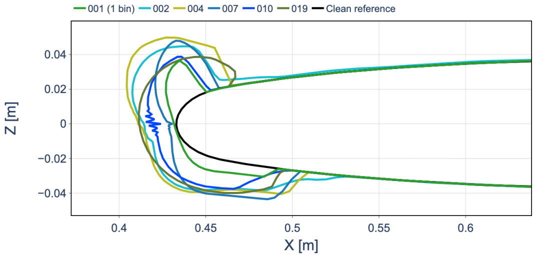
ONERA M6 – Ice Mass Roughness Sensitivity



Assessment

- One submission provided additional results for multiple prescribed roughness heights.
- Ice mass increases substantially with roughness, from approximately 1.3 kg at $k_s = 0.5$ mm to 1.9 kg at $k_s = 1.5$ mm on L1.

ONERA M6 – Ice Shapes (L1, 15 Bins)

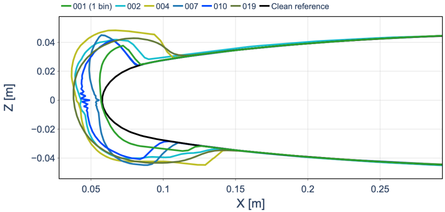


$y = 0.75$ **m – Roughness: 1 mm**

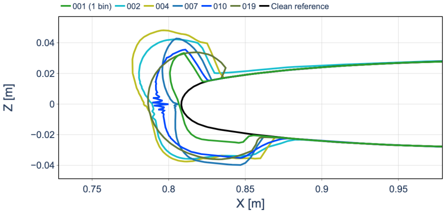
Assessment

- Considerable code-to-code variability is observed in the predicted ice shape, particularly in horn height, orientation, and local accretion thickness.
- Despite these differences, the submissions predict a broadly similar overall accretion region around the leading edge.
- Similar code-to-code variability is observed at the other spanwise locations.
- The geometric spread is consistent with the differences previously observed in the thermodynamic predictions, despite comparatively good agreement in water impingement.

ONERA M6 – Ice Shapes (L1, 15 Bins)



$y = 0.10$ **m**

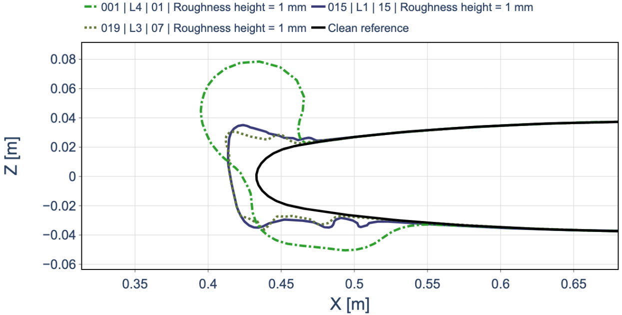


$y = 1.40$ **m**

Assessment

- Considerable code-to-code variability is observed in the predicted ice shape, particularly in horn height, orientation, and local accretion thickness.
- Despite these differences, the submissions predict a broadly similar overall accretion region around the leading edge.
- Similar code-to-code variability is observed at the other spanwise locations.
- The geometric spread is consistent with the differences previously observed in the thermodynamic predictions, despite comparatively good agreement in water impingement.

ONERA M6 – Multilayer Ice-Shape Comparison

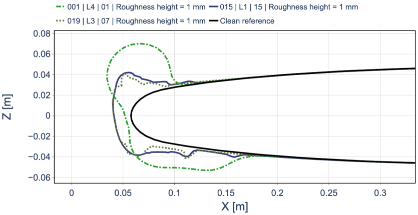


$y = 0.75$ **m**

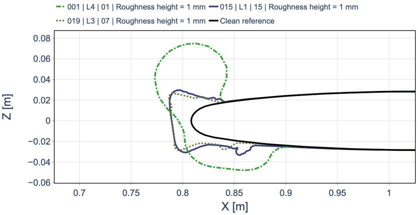
Assessment

- Considerable code-to-code variability is observed in horn orientation, extent, and local ice thickness.
- Predicted accretion ranges from relatively compact shapes to substantially larger upper- and lower-surface horns.
- Similar geometric variability persists at the inboard and outboard spanwise stations.

ONERA M6 – Multilayer Ice-Shape Comparison



$y = 0.10$ **m**



$y = 1.40$ **m**

Assessment

- Considerable code-to-code variability is observed in horn orientation, extent, and local ice thickness.
- Predicted accretion ranges from relatively compact shapes to substantially larger upper- and lower-surface horns.
- Similar geometric variability persists at the inboard and outboard spanwise stations.

ONERA M6 – Ice Accretion Summary

Ice Mass

- Grid sensitivity is submission-dependent, while sensitivity to droplet discretization remains limited.
- Code-to-code differences remain substantial and generally exceed the grid and droplet-discretization effects.

Mass Partition

- Most submissions convert approximately 85–89% of the impinged water mass into ice.
- Evaporation closes much of the remaining mass balance; lower ratios for some submissions are consistent with reported water shedding.

Ice Geometry

- Considerable code-to-code variability is observed in horn height, orientation, and local ice thickness.
- Similar geometric variability persists across the three spanwise sections.

Ice Accretion Takeaway

Good agreement in water impingement does not translate into comparable ice accretion; substantial differences emerge in mass partition and final ice geometry, highlighting the influence of the thermodynamic treatment.

ONERA M6 – Overall Assessment

Aerodynamics

- Limited grid and code-to-code variability is observed overall, except for C_D, which exhibits substantially larger differences.

Droplet Impingement

- Good overall agreement is observed, although impingement extent remains more sensitive to spatial and droplet-distribution resolution than integrated water mass.

Convective Heat Transfer & Thermodynamics

- Despite close agreement in T_{rec}, substantial code-to-code differences emerge in HTC, convective heat transfer, and particularly freezing fraction.

Ice Accretion

- Comparable water impingement does not translate into comparable ice accretion, with substantial variability in mass partition and final ice geometry.

Overall Takeaway

Code-to-code variability increases through the icing simulation chain, from comparatively consistent aerodynamics and impingement to substantially greater differences in thermodynamics and final ice accretion.